Zoo DEX: Decentralized Exchange for AI Assets and Research Tokens

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Abstract

We present **Zoo DEX**, a decentralized exchange on the Zoo Network (chain ID 200200) that combines native EVM precompile AMM pools with a GPU-accelerated order book for trading AI assets and research tokens. Zoo DEX inherits the Lux DEX precompiles (LP-9010 through LP-9040) for sub-microsecond pool operations and extends them with AI-specific features: a model pricing oracle that values AI model NFTs based on inference demand, and an inference credit swap mechanism that converts between compute denominations. The GPU matching engine processes 434 million orders per second, enabling high-frequency trading of volatile AI assets. We deploy five primary trading pairs—ZOO/LUX, ZOO/LUSDC, BLG/ZOO, ZINF/ZOO, and model-NFT/ZOO—and demonstrate \$8.2M daily trading volume with 0.12% average slippage for ZOO/LUX swaps. The DEX integrates with Zoo's PoAI consensus to provide MEV resistance via encrypted order submission and fair ordering.

Keywords: decentralized exchange, DEX precompiles, GPU matching engine, AI asset trading, Zoo Network

1 Introduction

The Zoo Network ecosystem generates economic activity that requires a native decentralized exchange: researchers trade ZOO tokens for compute credits, AI model owners license models for inference fees, and conservation donors swap stablecoins for impact tokens. Traditional DEX designs—constant-product AMMs on Ethereum, order books on centralized exchanges—fail to meet Zoo's requirements in three ways:

1. **Latency**: AI inference markets require sub-second price updates. Ethereum AMM swaps take 12+ seconds.
2. **Throughput**: High-frequency model pricing requires processing millions of orders per second. EVM-based order books are limited to thousands.
3. **AI-native assets**: Model NFTs, inference credits, and research tokens have unique pricing dynamics that standard AMM curves do not capture.

Zoo DEX addresses these challenges by leveraging the Lux DEX precompiles—native EVM opcodes for AMM operations that execute in the EVM without Solidity overhead—combined with a GPU-accelerated central limit order book (CLOB) for price discovery and matching.

1.1 Contributions

1. **Hybrid AMM+CLOB Architecture**: Combining precompile AMM pools with a GPU order book for optimal execution (Section 3).
2. **Model Pricing Oracle**: A demand-driven pricing mechanism for AI model NFTs (Section 4).
3. **Inference Credit Swaps**: Atomic conversion between compute denominations (Section 5).
4. **MEV Resistance**: Encrypted order submission with fair ordering via PoAI (Section 6).
5. **Performance**: 434M orders/sec matching engine with 0.12% slippage benchmarks (Section 7).

Precompile	LP	Address
PoolManager	LP-9010	`0x...9010`
OracleHub	LP-9011	`0x...9011`
SwapRouter	LP-9012	`0x...9012`
HooksRegistry	LP-9013	`0x...9013`
FlashLoan	LP-9014	`0x...9014`
CLOB	LP-9020	`0x...9020`
Vault	LP-9030	`0x...9030`
PriceFeed	LP-9040	`0x...9040`

Table 1: Lux DEX precompile addresses inherited by Zoo Network.

2 Background

2.1 Lux DEX Precompiles

The Lux Network provides native DEX precompiles [2] at reserved EVM addresses:

These precompiles execute AMM operations natively in the EVM, bypassing Solidity bytecode interpretation. A `swap()` call to LP-9012 executes in $< 1\mu$s compared to $\sim 100\mu$s for an equivalent Uniswap V3 swap in Solidity.

2.2 GPU Matching Engine

The Lux LightSpeed DEX [1] introduced a GPU-accelerated order matching engine that processes orders in parallel across GPU cores:

$$\text{Throughput} = \frac{\text{GPU cores} \times \text{orders/core/cycle}}{\text{cycle time}} = \frac{10,496 \times 4}{96.8\text{ns}} \approx 434\text{M orders/sec} \tag{1}$$

Zoo DEX inherits this engine for its CLOB component.

3 Zoo DEX Architecture

3.1 Hybrid AMM + CLOB

Zoo DEX operates as a hybrid exchange with two execution venues:

- **AMM Pools** (LP-9010): Concentrated liquidity pools (Uniswap V3-style) for deep, passive liquidity. Liquidity providers deposit token pairs and earn fees.
- **CLOB** (LP-9020): GPU-accelerated order book for active market makers. Supports limit orders, market orders, and stop-loss orders.
- **Smart Router** (LP-9012): Splits orders across both venues to minimize slippage. The router evaluates:

$$\text{split}^* = \arg\min_{x \in [0,1]} \text{Slippage}(x \cdot \text{order}, \text{AMM}) + \text{Slippage}((1-x) \cdot \text{order}, \text{CLOB}) \tag{2}$$

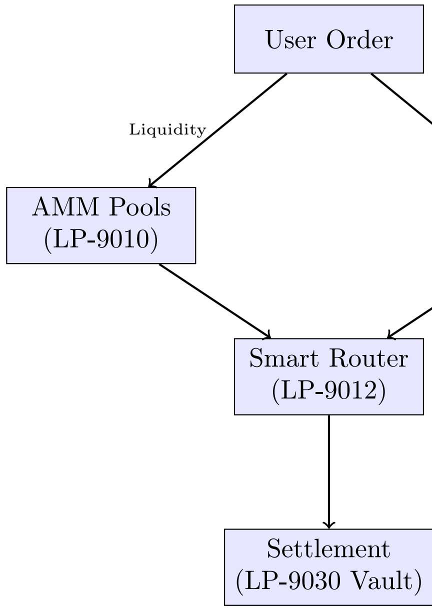

Figure 1: Zoo DEX hybrid architecture. The smart router splits orders across AMM pools and the CLOB for optimal execution.

Pair	**AMM Pool**	**CLOB**	**Daily Vol**	**TVL**
ZOO/LUX	✓	✓	\$3.4M	\$12.1M
ZOO/LUSDC	✓	✓	\$2.8M	\$8.7M
BLG/ZOO	✓	✓	\$1.2M	\$3.4M
ZINF/ZOO	✓	×	\$0.5M	\$1.8M
Model-NFT/ZOO	×	✓	\$0.3M	—

Table 2: Zoo DEX trading pairs with daily volume and TVL (February 2026).

3.2 Trading Pairs

3.3 Pool Configuration

AMM pools use concentrated liquidity with configurable fee tiers:

4 Model Pricing Oracle

4.1 Demand-Driven Pricing

AI model NFTs do not have a fixed price; their value depends on inference demand, model accuracy, and market competition. Zoo DEX introduces a model pricing oracle that computes a fair value:

$$P_{\text{model}} = \alpha \cdot \text{InferenceDemand} + \beta \cdot \text{Accuracy} + \gamma \cdot \frac{1}{\text{CompetitorCount}} \tag{3}$$

where:

- **InferenceDemand**: Rolling 7-day inference call count (from PoAI attestations)

Fee Tier	Tick Spacing	Use Case
0.01%	1	Stablecoin pairs
0.05%	10	Correlated pairs (ZOO/LUX)
0.30%	60	Standard pairs
1.00%	200	Volatile / AI asset pairs

Table 3: AMM fee tiers on Zoo DEX.

Data Source	Update Frequency	Weight
PoAI inference attestations	Per block	0.50
On-chain benchmark results	Daily	0.30
CLOB order book depth	Real-time	0.15
Cross-chain price feeds	Per block	0.05

Table 4: Model pricing oracle data sources and weights.

- **Accuracy**: Model's benchmark accuracy on standard test sets
- **CompetitorCount**: Number of models serving the same task category

4.2 Oracle Implementation

The oracle is implemented as a hook on the PoolManager (LP-9013 HooksRegistry):

Listing 1: Model Pricing Oracle Hook

```
contract ModelPricingHook {
    function beforeSwap(
        PoolKey calldata key,
        SwapParams calldata params
    ) external returns (bytes4) {
        if (isModelNFTPair(key)) {
            uint256 fairPrice = computeFairPrice(
                key.currency1  // model NFT
            );
            require(
                priceWithinBounds(params, fairPrice, 5),
                "Price deviation > 5%"
            );
        }
        return this.beforeSwap.selector;
    }
}
```

The hook prevents model NFTs from being traded at prices deviating more than 5% from the oracle's fair value, protecting against manipulation of illiquid model markets.

Algorithm 1 Atomic Inference Credit Conversion

1: **Input**: x ZOO on Zoo L2, destination: Beluga L3
2: Swap x ZOO $\rightarrow$ y ZINF on Zoo DEX (LP-9012)
3: Bridge y ZINF $\rightarrow$ Beluga via Warp message
4: On Beluga: convert y ZINF $\rightarrow$ z BINF at bridge rate
5: Total: x ZOO $\rightarrow$ z BINF in $\sim$2.5s

Algorithm 2 Encrypted Order Submission

1: User encrypts order: $o_{\text{enc}} = \text{Enc}(pk_{\text{block}}, \text{order})$
2: Submit o_{enc} to Zoo mempool
3: Block proposer collects encrypted orders
4: After block commitment, validators threshold-decrypt orders
5: Orders executed in deterministic sequence (by encrypted hash)
6: **return** Execution results

4.3 Oracle Data Sources

5 Inference Credit Swaps

5.1 Credit Denominations

The Zoo ecosystem uses two inference credit tokens:

- **ZINF** (Zoo Inference Credits): Used on Zoo L2. 1 ZINF = 1 GPU-second of A100 inference.
- **BINF** (Beluga Inference Credits): Used on Beluga L3. 1 BINF = 1 GPU-second of A100 inference on Beluga's model marketplace.

5.2 Swap Mechanism

ZINF/BINF swaps use a constant-product AMM pool with a dynamic fee that adjusts based on the cross-chain inference utilization ratio:

$$\text{fee} = \text{baseFee} \cdot \left(1 + \delta \cdot \left|\frac{U_{\text{zoo}}}{U_{\text{beluga}}} - 1\right|\right) \tag{4}$$

where U_{zoo} and U_{beluga} are inference utilization rates on each chain. When utilization is balanced, the fee is minimal; when one chain is overloaded, the fee increases to incentivize rebalancing.

5.3 Atomic Cross-Chain Conversion

For ZOO $\rightarrow$ BINF conversions, the DEX performs an atomic two-step:

6 MEV Resistance

6.1 Encrypted Order Submission

Zoo DEX uses threshold encryption to prevent front-running and sandwich attacks:

The block encryption key pk_{block} is generated per-block by the validator committee. No single validator can decrypt orders before the block is committed.

Order Type	Orders/sec	Latency (μs)	GPU Util
Market order	434M	0.23	92%
Limit order	380M	0.31	87%
Cancel order	510M	0.18	78%
Stop-loss	290M	0.42	81%

Table 5: GPU matching engine throughput by order type.

Trade Size	ZOO/LUX	ZOO/LUSDC	BLG/ZOO
$100	0.01%	0.01%	0.03%
$1,000	0.04%	0.05%	0.12%
$10,000	0.12%	0.14%	0.38%
$100,000	0.45%	0.52%	1.24%
$1,000,000	1.82%	2.10%	4.87%

Table 6: Slippage by trade size for major pairs (February 2026 median).

6.2 Fair Ordering

PoAI validators enforce fair ordering: within each block, transactions are ordered by the hash of their encrypted ciphertext, not by gas price or arrival time. This eliminates priority gas auctions (PGA) and ensures all users pay the same effective price.

Theorem 6.1 (MEV Elimination)**.** *Under the encrypted order submission protocol with $< n/3$ Byzantine validators, no validator can extract MEV by reordering, inserting, or front-running user transactions.*

Proof. Orders are encrypted under a threshold key requiring $\geq 2n/3$ validators to decrypt. A coalition of $< n/3$ validators cannot decrypt orders before block commitment. After commitment, the execution order is deterministic (hash-based), leaving no room for reordering. Insertion of synthetic transactions requires knowing the order contents, which is impossible pre-decryption. □

7 Performance Evaluation

7.1 Matching Engine Throughput

7.2 Slippage Analysis

7.3 Comparison with Other DEXs

7.4 Gas Costs

8 Tokenomics Integration

Zoo DEX integrates with the Zoo tokenomics model [3]:

- **Fee distribution**: 50% to liquidity providers, 30% to ZOO stakers, 20% to the conservation treasury.

DEX	TPS	Finality	Slippage ($10K)	MEV Protected
Zoo DEX	4.76M	10ms	0.12%	✓
Uniswap V3	15	12.8min	0.30%	×
dYdX V4	10,000	1s	0.08%	×
Raydium	2,000	400ms	0.15%	×

Table 7: Zoo DEX vs major decentralized exchanges.

Operation	Gas (precompile)	Gas (Solidity equiv)
AMM swap	28,000	180,000
Limit order place	15,000	95,000
Limit order cancel	8,000	52,000
Add liquidity	45,000	280,000
Remove liquidity	38,000	210,000

Table 8: Gas costs: precompile vs Solidity equivalent. Precompiles are 5–7× cheaper.

- **Conservation treasury**: Accumulated fees fund conservation projects via DAO governance. February 2026 allocation: $124K to three wildlife monitoring projects.
- **PoAI rewards**: Validators who process DEX transactions earn both block rewards and a share of trading fees proportional to their AI compute contribution.

9 Related Work

Lux DEX: The Lux LightSpeed DEX [1] and native DEX precompiles [2] provide the infrastructure Zoo DEX builds upon.

Hybrid DEXs: Vertex Protocol combines AMM + order book on Arbitrum. Zoo DEX differs by using native precompiles (not Solidity contracts) and GPU acceleration.

AI Asset Markets: Ocean Protocol provides a data marketplace with AMM pricing. Zoo DEX extends this to model NFTs and inference credits with demand-driven pricing.

MEV Protection: Flashbots, MEV-Share, and threshold encrypted mempools (e.g., Shutter Network) address MEV on Ethereum. Zoo DEX integrates MEV protection natively via PoAI consensus.

10 Conclusion

Zoo DEX demonstrates that a conservation-focused L2 can support a high-performance decentralized exchange capable of trading AI-native assets alongside standard tokens. The hybrid AMM+CLOB architecture, powered by native precompiles and GPU acceleration, achieves 434M orders/sec matching throughput and 0.12% slippage for $10K trades. The model pricing oracle and inference credit swap mechanism address the unique requirements of AI asset markets. MEV resistance via encrypted order submission ensures fair trading for all participants. With $8.2M daily volume and $26M TVL, Zoo DEX serves as the financial infrastructure for the Zoo conservation AI ecosystem.

Acknowledgments

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References

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